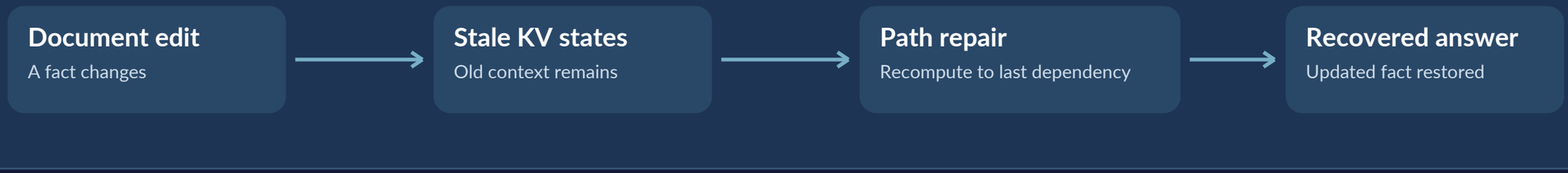
RESEARCH PRESENTATION

How Far Does an Edit Reach?

Repairing Derived Facts in Stale KV Caches

THE RESEARCH IDEA



PRESENTED BY

Nour Shreif • Hadi Ghazi • Karin Al Zoabi

Lebanese University

2026

THE PROBLEM + THE IDEA

How far does an edit reach?

Repairing derived facts in stale KV caches

Why a one-sentence edit can break later answers

01 ORIGINAL DOCUMENT

Varnex uses DELTA

Delta maps to code Q7

Cached answer

Q7

02 DOCUMENT EDIT

Varnex now uses LAMBDA

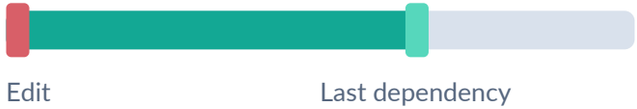
Lambda maps to code R6

Correct answer

R6

03 OUR SOLUTION

Recompute only the
affected downstream path



Avoid re-reading the tail

CENTRAL QUESTION

How can we identify the **last sentence affected by an edit** and repair its cached dependencies **without recomputing the entire document?**

METHOD + EXPERIMENT

From edit to repaired cache

Many repair policies | three 7B–8B models | controlled WikiText-103 benchmark

1 **EDIT** Cache the document, then change one fact → 2 **FIND** Sentences affected by the edit → 3 **BOUNDARY** The last affected sentence → 4 **RECOMPUTE** Contiguous KV path: edit → boundary → 5 **EVALUATE** New answer recovered? Tokens recomputed?

FINDING THE BOUNDARY: TWO FINDERS

Both read only the text, never the cache, and return a stopping sentence.

FAST **1 Word index** no model, every edit

Words the edit changed: **Delta, Lambda** → Mark every later sentence containing either → Follow rare names & codes, up to **3 hops**

Both words matter: the Lambda row is newly relevant, but the Delta row is stale too.

SEMANTIC **2 GPT-4.1, for paraphrase** "the third plan" ↔ "Plan three"

ONCE, WHEN STORED
Builds a dependency graph → **old links**

AT EDIT TIME
Asked about the new version → **new links**

Stop at the furthest sentence either one reaches.

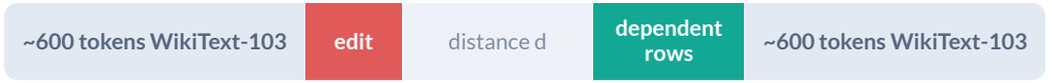
REPAIR POLICIES · 3 INSTRUCTION-TUNED MODELS

Qwen3-8B | Llama-3.1-8B | Mistral-7B

No repair (floor) · Full re-read (ceiling) · Mao: fixed windows · Mao: CarrierWindow · Li: appended correction · … and more

CONTROLLED BENCHMARK

9 settings × 50 documents = 450 / model



Direct · **derived** at d = 0 / 250 / 1,500 · **3-hop** · **fan-out** · **noisy** · **paraphrased** at d = 250 / 1,500

Rows in random order · every edit keeps the same token length in all three tokenizers

MARGIN RECOVERY **0** = no better than stale cache — **1** = as good as full re-read + answer accuracy + recomputed tokens

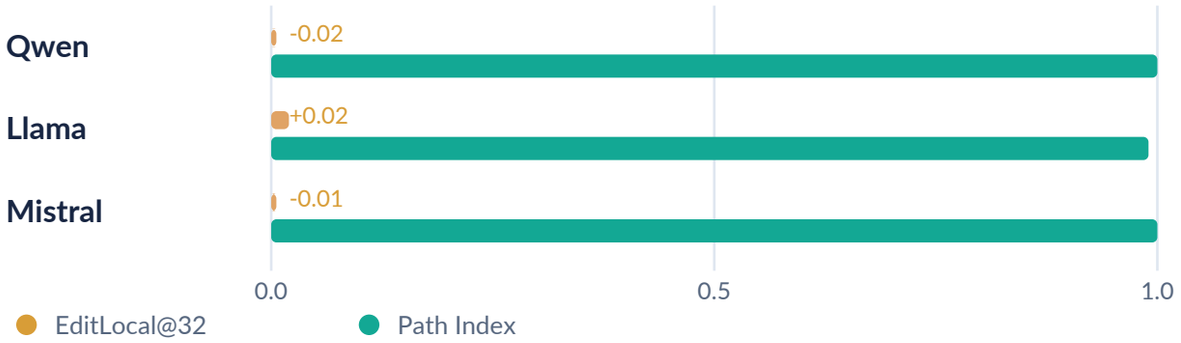
RESULTS + DEFENSE TAKEAWAY

What the experiments establish

Strong answer recovery when the correct repair endpoint is identified

DISTANT DERIVED DEPENDENCIES | d = 1,500

Margin recovery: EditLocal@32 vs. Path Index



EXPLICIT DEPENDENCIES

0.99-1.00

margin recovery (Path Index)

95-100%

answer accuracy on retained cases

Across 18 derived setting × model cells

RECOMPUTED TOKENS VS FULL RE-READ

d = 250

18% of full (368 / 2,008)

d = 1,500

50% of full (1,628 / 3,257)

PARAPHRASED DEPENDENCIES

Word index

0.11-0.42 MR

GPT-4.1, combined

0.84-0.98 MR

Attention@512: 0.98-1.03 MR (extra overhead excluded)